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# NumPy Filter Array

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## Filtering Arrays

Getting some elements out of an existing array and creating a new array out of them is called *filtering*.

In NumPy, you filter an array using a *boolean index list*.

A *boolean index list* is a list of booleans corresponding to indexes in the array.

If the value at an index is **True** that element is contained in the filtered array, if the value at that index is **False** that element is excluded from the filtered array.

## Example

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Create an array from the elements on index 0 and 2:

```
import numpy as np

arr = np.array([41, 42, 43, 44])

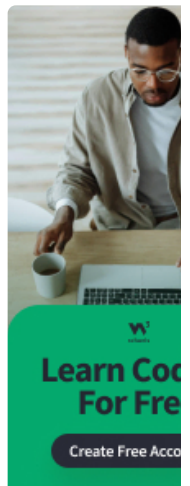
x = [True, False, True, False]

newarr = arr[x]

print(newarr)
```

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The example above will return `[41, 43]`, why?

Because the new array contains only the values where the filter array had the value `True`, in this case, index 0 and 2.



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## Creating the Filter Array

In the example above we hard-coded the `True` and `False` values, but the common use is to create a filter array based on conditions.

### Example

Create a filter array that will return only values higher than 42:

```
import numpy as np

arr = np.array([41, 42, 43, 44])

# Create an empty list
filter_arr = []

# go through each element in arr
for element in arr:
    # if the element is higher than 42, set the value
    # to True, otherwise False:
    if element > 42:
        filter_arr.append(True)
    else:
        filter_arr.append(False)

newarr = arr[filter_arr]

print(filter_arr)
print(newarr)
```

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## Example

Create a filter array that will return only even elements from the original array:

```
import numpy as np

arr = np.array([1, 2, 3, 4, 5, 6, 7])

# Create an empty list
filter_arr = []

# go through each element in arr
for element in arr:
    # if the element is completely divisible by 2, set
    # the value to True, otherwise False
    if element % 2 == 0:
        filter_arr.append(True)
    else:
        filter_arr.append(False)

newarr = arr[filter_arr]

print(filter_arr)
print(newarr)
```

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## Creating Filter Directly From Array

The above example is quite a common task in NumPy and NumPy provides a nice way to tackle it.

We can directly substitute the array instead of the iterable variable in our condition and it will work just as we expect it to.

## Example

Create a filter array that will return only values higher than 42:

```
import numpy as np

arr = np.array([41, 42, 43, 44])

filter_arr = arr > 42

newarr = arr[filter_arr]
```

```
print(filter_arr)
print(newarr)
```

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## Example

Create a filter array that will return only even elements from the original array:

```
import numpy as np

arr = np.array([1, 2, 3, 4, 5, 6, 7])

filter_arr = arr % 2 == 0

newarr = arr[filter_arr]

print(filter_arr)
print(newarr)
```

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## Exercise ?

Consider the following code:

```
import numpy as np
arr = np.array(['a', 'b', 'c'])
x = arr[[True, False, True]]
print(x)
```

What will be the printed result?

- ☐ ['b']
- ☐ ['a' 'c']
- ☐ ['a' 'b' 'c']

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